

MAXELL G. MILAY

AI Engineer | LLM Engineer | Prompt Engineer
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PROFESSIONAL SUMMARY

AI Engineer with 3+ years architecting, deploying, and scaling production-grade Generative AI and Large Language Model (LLM) systems for enterprise clients across insurance, BPO, and SaaS. Specialized in LLM orchestration, prompt engineering, multi-agent workflows with LangChain and LangGraph, Retrieval-Augmented Generation (RAG) pipelines, vector databases, and MLOps on AWS and GCP. Proven impact reducing LLM API costs 30-40%, improving extraction accuracy from 78% to 92%, and shipping AI features end-to-end across the full ML lifecycle. Skilled in optimizing prompts, context windows, and model selection for cost-efficient, high-quality LLM applications.

CORE SKILLS

LLM and Generative AI: OpenAI API, GPT-4, GPT-4o, GPT-4o-mini, Prompt Engineering, Prompt Optimization, Chain-of-Thought Prompting, Few-Shot Prompting, System Prompts, Context Window Optimization, Response Caching, LangChain, LangGraph, LlamaIndex, Multi-Agent Systems, Agentic AI, Agentic Workflows, AI Orchestration, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Semantic Search, Vector Databases, pgvector, OpenAI Embeddings, Function Calling, Tool Use, Hugging Face Transformers

Machine Learning and NLP: TensorFlow, Keras, PyTorch, Scikit-learn, spaCy, Hugging Face, Named Entity Recognition (NER), Sentiment Analysis, Transformer Models, CNN, RNN, LSTM, K-Means, Random Forest, XGBoost, CatBoost, NSGA-II Genetic Algorithms, Reinforcement Learning, Deep Q-Learning, Proximal Policy Optimization (PPO), Monte Carlo Simulation, Feature Engineering, Hyperparameter Tuning, Model Evaluation, A/B Testing, Intelligent Document Processing (IDP)

MLOps, Cloud, and Engineering: AWS Lambda, Step Functions, Amazon Textract, S3, DynamoDB, CloudWatch, Google Cloud Platform (GCP), Vertex AI, Cloud Run, Cloud SQL, Terraform (IaC), Docker, GitHub Actions, CI/CD, Continuous Training (CT), MLflow, Model Monitoring, AI Observability, Token Usage Optimization, Python, JavaScript, TypeScript, SQL, Django REST Framework, FastAPI, Vue.js, React, Streamlit, Plotly, PostgreSQL, REST APIs, ETL Pipelines

PROFESSIONAL EXPERIENCE

AI Engineer - ECLARO

Quezon City, Philippines (Remote) | April 2025 - Present

- Architected enterprise-scale Generative AI orchestration framework using OpenAI GPT-4 and GPT-4o with AWS Lambda and Step Functions, enabling secure LLM deployment across 5+ insurance business units processing 10,000+ monthly documents.
- Pioneered multi-agent workflow prototypes with LangChain and LangGraph, achieving 40% reduction in underwriting processing time and 35% improvement in decision consistency across test scenarios.
- Championed Model Context Protocol (MCP) adoption by developing 6+ custom connectors integrating internal policy APIs, claims databases, and document repositories for secure, contextual AI assistants.
- Optimized LLM inference costs by 30% through advanced prompt engineering, response caching, and model selection strategies (GPT-4 vs GPT-4o-mini) while maintaining 90%+ response quality benchmarks.
- Redesigned Intelligent Document Processing (IDP) pipeline using Amazon Textract, custom Named Entity Recognition (NER) models, and spaCy NLP, improving field-level extraction accuracy from 78% to 92% across complex insurance forms (including FNOL claims) and reducing manual review by 45%.
- Implemented AI observability stack with AWS CloudWatch and custom metric dashboards, achieving 99.5% framework uptime and reducing incident detection time by 60%.

Lead Software Engineer - BitWork Solutions

Miami, FL, USA (Remote) | December 2024 - April 2025

- Led cross-functional team of 5 engineers delivering an image-processing mobile application from concept to production, achieving 95% sprint velocity across 8+ development cycles.
- Spearheaded backend migration from Directus headless CMS and AWS Lambda to a scalable Django REST Framework and PostgreSQL architecture, cutting infrastructure cost 35% and improving API response times 50%.
- Implemented CI/CD pipelines with automated testing and deployment workflows, reducing release cycle time from 2 weeks to 3 days.
- Orchestrated agile sprint planning and backlog prioritization for 50+ user stories, maintaining 90% on-time delivery and reducing scope creep 40%.

DevOps Engineer (Part-time) - BitWork Solutions

Miami, FL, USA (Remote) | July 2024 - December 2024

- Architected CI/CD pipelines with automated linting and testing across 3 enterprise projects, reducing manual QA effort 70% and catching 85% of bugs pre-production.
- Engineered AWS infrastructure (S3, DynamoDB, Lambda) projected to support 50,000+ daily image uploads at 99.9% uptime.
- Re-architected legacy backend to serverless AWS Lambda functions, achieving 60% server cost reduction and elastic scaling from 100 to 5,000 requests per minute.
- Implemented Infrastructure-as-Code with Terraform, reducing environment provisioning time from 4 hours to 15 minutes.

AI Engineer - BPOSeats

Mandaue City, Philippines | June 2024 - January 2025

- Architected ML-powered productivity forecasting system processing 15,000+ historical task records, projected to lift operational efficiency 55% and streamline resource allocation.
- Engineered prompt engineering strategies for OpenAI GPT models and built an enterprise LLM cost monitoring dashboard tracking token usage across 8+ features, reducing operational expense 35-40% via context window optimization, response caching, and model selection.
- Co-led AI-driven effort estimation framework integrating NSGA-II genetic algorithms with LLM reasoning, improving task duration prediction accuracy 30% over baseline methods.
- Designed real-time AI-spending dashboards with Python and Plotly to enable data-driven AI feature prioritization.

Full Stack Web Developer - BPOSeats

Mandaue City, Philippines | September 2023 - May 2024

- Pioneered company's first production-grade vector database with PostgreSQL and pgvector, enabling semantic search across 5,000+ documentation entries with 85% retrieval accuracy.
- Developed RAG-powered customer support chatbot leveraging OpenAI embeddings and semantic search, projected to deflect 75% of tier-1 tickets and reduce average resolution time 40%.
- Engineered automated ETL pipeline migrating 2,500+ legacy documentation pages to vector database with chunking and embedding strategies, achieving sub-3-second response times.
- Architected testing infrastructure integrating Vue.js and Django unit tests into CI/CD pipelines, reducing production deployment failures 80% and lifting system uptime from 94% to 99%+.

Web Developer Intern - Symph

Cebu City, Philippines | December 2022 - March 2023

- Deployed code changes to 3+ client websites with backend and DevOps teams, learning production deployment workflows and staging-environment best practices.
- Diagnosed and resolved 15+ UI/UX issues across 2 active client projects, improving page load performance 25% and mobile responsiveness across 5+ key user flows.
- Implemented client-requested feature enhancements and bug fixes using HTML, CSS, JavaScript, and React, contributing to 20% increase in client satisfaction.

SELECTED PROJECTS

Beacon - AI Career Roadmap Generator

1st Place, UP Computer Science Guild Komsai Week Hackathon 2024

- Led 5-person team to ship working prototype in 6 hours; generative AI produces personalized visual career roadmaps from user profile data.
- Implemented authentication and storage with Google Firebase; integrated OpenAI LLM models for roadmap generation.

Lakbai - GIS Pedestrian Accessibility Platform

Champion, 2024 Philippine Junior Data Science Challenge

- Built data-driven pedestrian accessibility index powered by a fuzzy logic system and multilayer perceptron in 2 weeks.
- Deployed frontend on Vercel, backend on Google Cloud Run, and PostgreSQL on Cloud SQL; built CI/CD pipelines with GitHub Actions tying GCP services together.

Finite Automaton Visualizer

Interactive web app for regex-to-DFA visualization

- Led 4-person team building interactive web app that generates DFA graphs from regular expressions and visually simulates string inclusion.
- Implemented core algorithm for abstract syntax tree generation from regex with operator-precedence handling.

VOLUNTARY EXPERIENCE

AI Infrastructure and Deployment Advocate - AI Pilipinas - Cebu Chapter

Cebu City, Philippines | July 2024 - Present

- Researched and disseminated MLOps best practices across the full ML lifecycle (data ingestion, feature engineering, training, deployment, monitoring) and authored Medium articles on Vertex AI and GCP for the PH AI community.

Lead Software Engineer - University of the Philippines Computer Science Guild

Cebu City, Philippines | November 2023 - Present

- Led 7 designers and 10 developers shipping multiple versions of the official UP CS Guild website; established designer-developer agile workflow and deployed via Docker and GitHub Actions on university servers.

EDUCATION

Bachelor of Science in Computer Science

University of the Philippines Cebu | Cebu City, Philippines

- Led 3 teams through small- and large-scale competitions and projects, achieving top placements and highest ratings.

CERTIFICATIONS

Machine Learning Specialization - DeepLearning.AI and Stanford University - Supervised, unsupervised, and reinforcement learning with TensorFlow and Scikit-learn; built Deep Q-Learning Lunar Lander and recommender systems.

Deep Learning Specialization - DeepLearning.AI and Stanford University - Built industry-grade CNNs and RNNs including VGG-16/19 and MobileNet variants on CIFAR-10.